GTIIT Spring 2025 Calculus 1M
Workshop 13

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

Exercise 1. For what values of p is each series convergent?

- $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$
- $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{(\ln n)^p}{n}$

Exercise 2. Use the Ration Test to determine whether the series is convergent or divergent:

- $\sum_{n=1}^{\infty} \frac{n!}{n^n}$
- $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}$

Exercise 3. Test the series for convergence or divergence:

- $\sum_{n=1}^{\infty} \frac{n!}{e^{n^2}}$
- $\sum_{j=1}^{\infty} \frac{\sqrt{j}}{j+5}$

Exercise 4. Find the radius of convergence and interval of convergence of the series.

- $\sum_{n=1}^{\infty} \frac{n}{2^n(n^2+1)} x^n$
- $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{5^n \sqrt{n}}$

Exercise 5. Show that the function

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

satisfies the equation:

$$f''(x) + f(x) = 0.$$